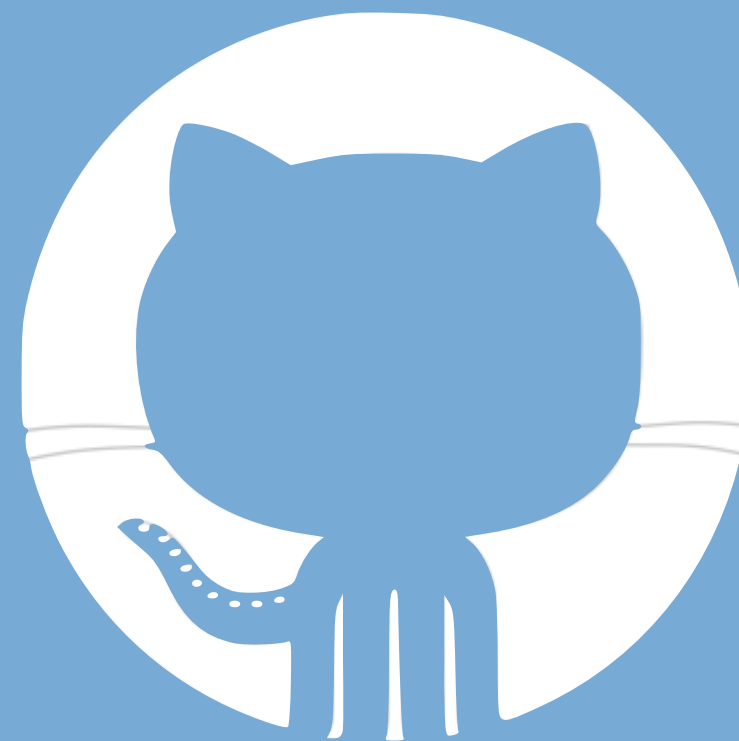


CRDTs

Data Types For EC Systems

joeljacobson



basho.com

PROBLEM?

Why even is there CRDTs?

Eventual Consistency

“Eventual consistency is a consistency model used in distributed computing that informally guarantees that, if no new updates are made to a given data item, eventually all accesses to that item will return the last updated value.”

--Wikipedia

Trade Off

EC is a trade off, no one _wants_ EC,
they want what it buys them.

Scale Up!

\$\$\$Big Iron
(still fails)

Scale Out

Commodity Servers

CDNs, App Servers

\$\$ Expertise

Fault Tolerance

Still works when things fail
(‘cos things fail)

Low Latency

“Amazon found every 100ms of latency
cost them 1% in sales”

Low Latency

“Google found an extra 0.5 seconds in search page generation time dropped traffic by 20%.”

CAP

CAP

PACELC

Trading C for LL / A

PACELC

Trading C for LL / A

PACELC

Trading C for LL / A

EC

Causal

RYOW

Session

Monotonic Read / Write

PICK YOUR OWN

<http://basho.com/understanding-riaks-configurable-behaviors-part-1/>

<http://www.bailis.org/blog/when-is-acid-acid-rarely/>

WHO PAYS?

DEVELOPER

But how?

Google F1

"We...have a lot of experience with eventual consistency systems at Google...we find developers spend a significant fraction of their time building extremely complex and error-prone mechanisms to cope with eventual consistency"

Riak Overview

Riak Overview

Erlang implementation of Dynamo

Riak Object

Key

Value

Riak Overview

Consistent hashing

Riak Overview

Dynamic membership

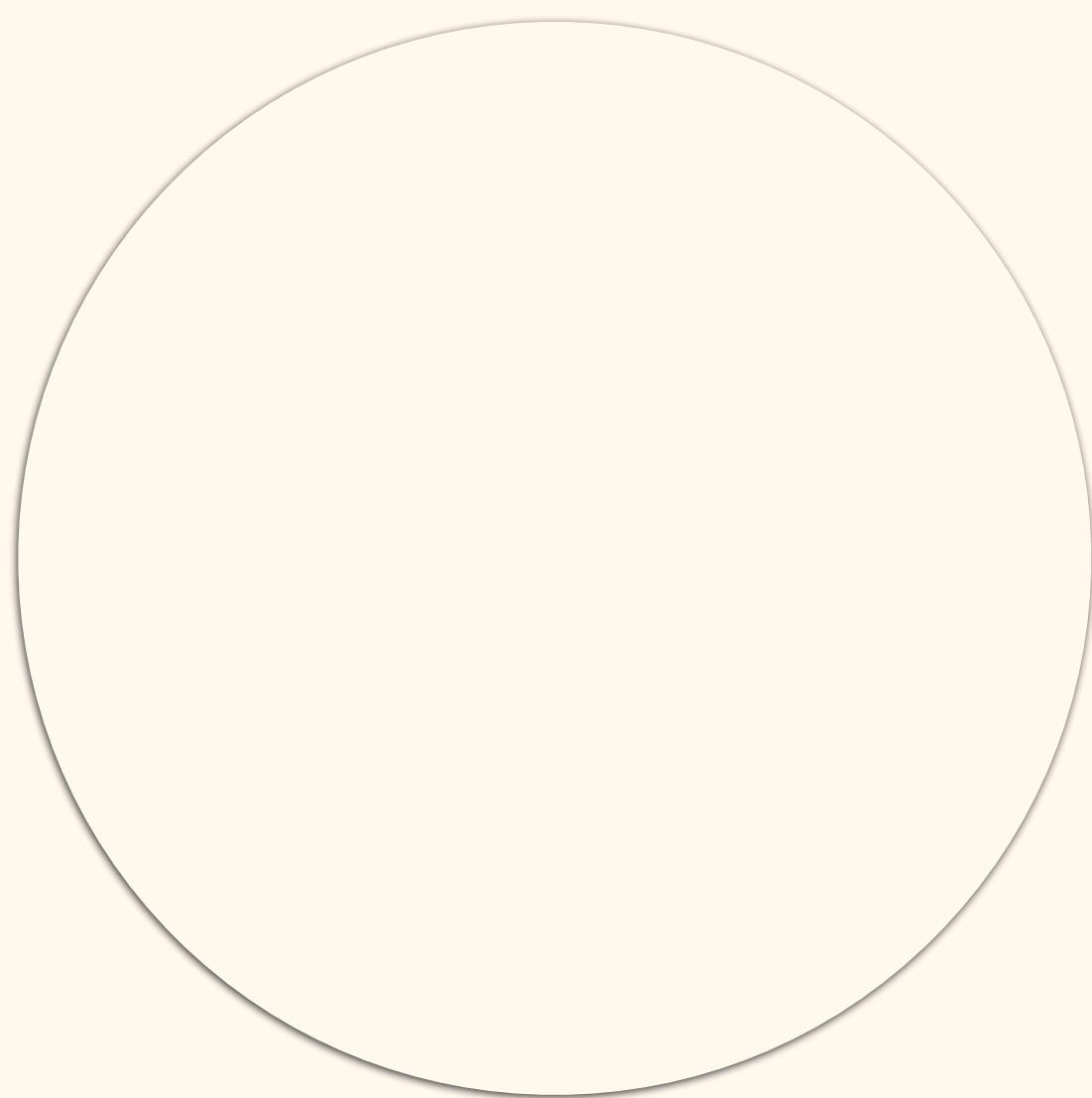
Riak Overview

Replication factor

Replica

Replica

Replica



High Availability

“...any non-failing node can respond to any request”

--Gilbert & Lynch

Riak Overview

Two Writes: {Writer, Value, Time}

$\{a, v1, a1\}$

$\{b, v2, b1\}$

$\{a, v1, a1\}$

Riak Overview

Last Writer Wins

Allow Mult

Riak Overview

Last Writer Wins

$\{b, v1, t2\}$

$\{b, v1, t2\}$

$\{b, v1, t2\}$

Riak Overview

Allow Mult

$[\{a, v1, a1\}, \{b, v2, b1\}]$

$[\{a, v1, a1\}, \{b, v2, b1\}]$

$[\{a, v1, a1\}, \{b, v2, b1\}]$

User specified Merge

SEMANTIC RECONCILIATIO N

DB IS DOWN

Fill in Form P17QR-35

File for input

DB IS DOWN

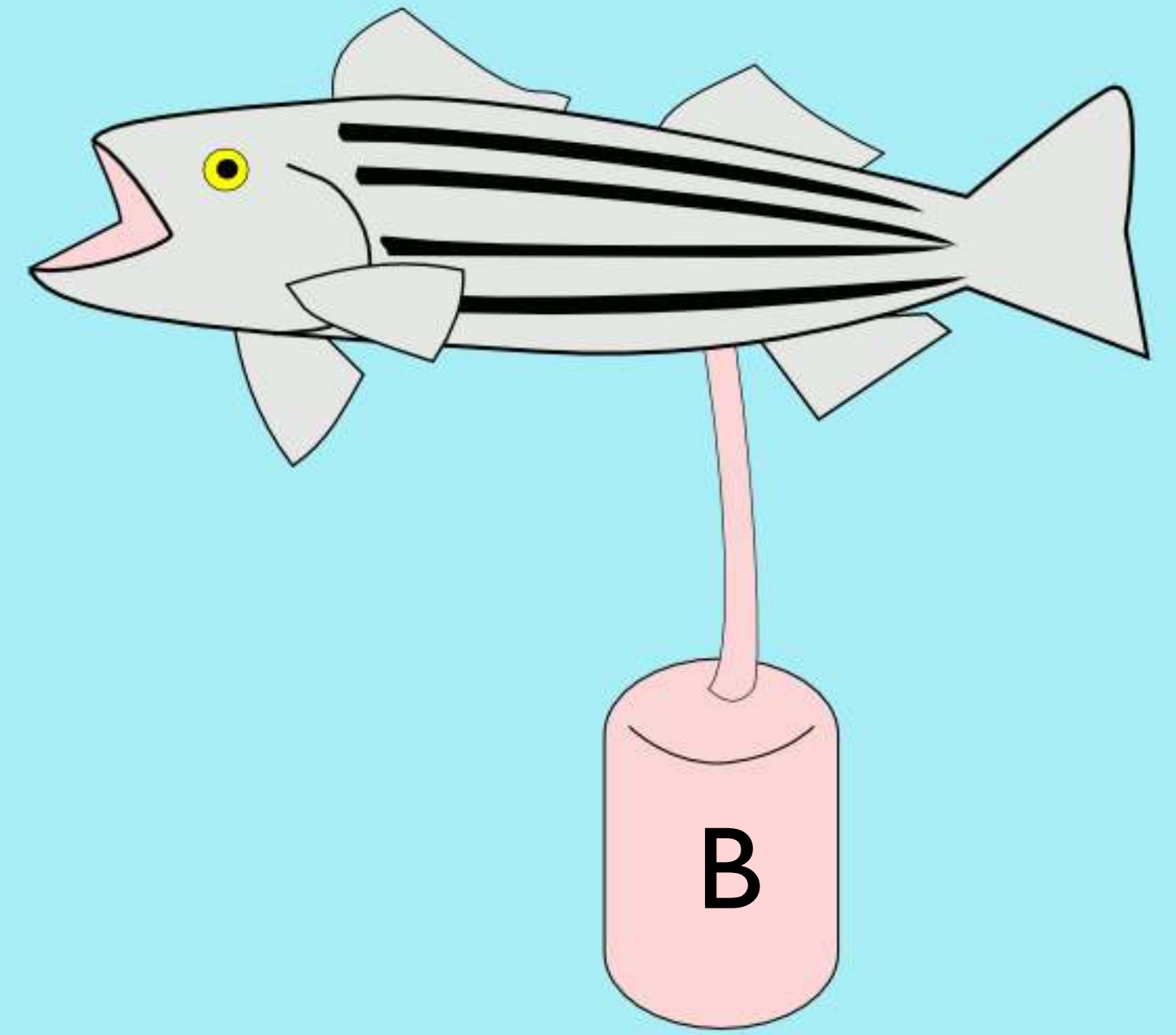
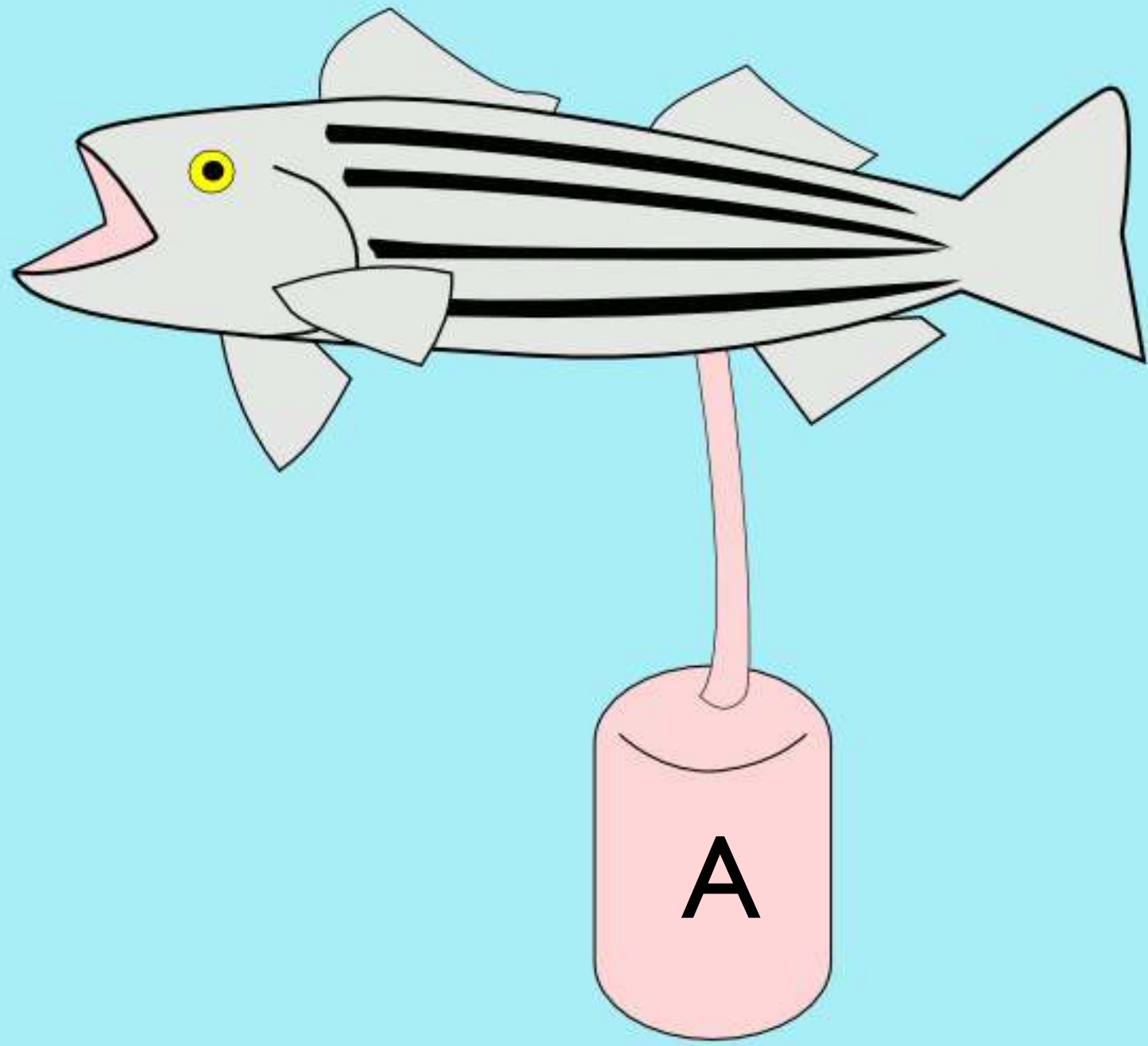
Fill in Form P17QR-35

File for input

OOPS

DYNAMO

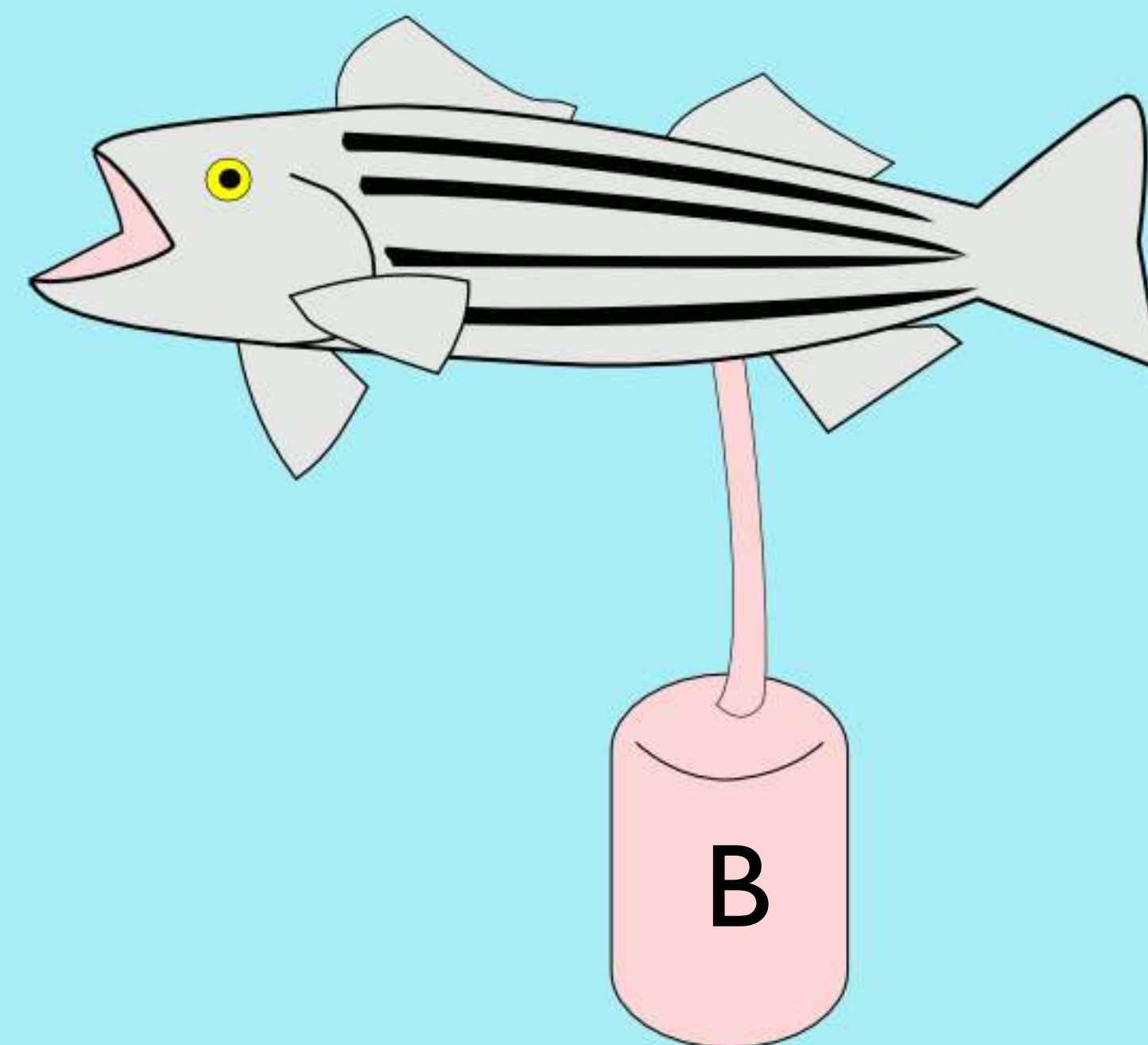
The Shopping Cart Part.1



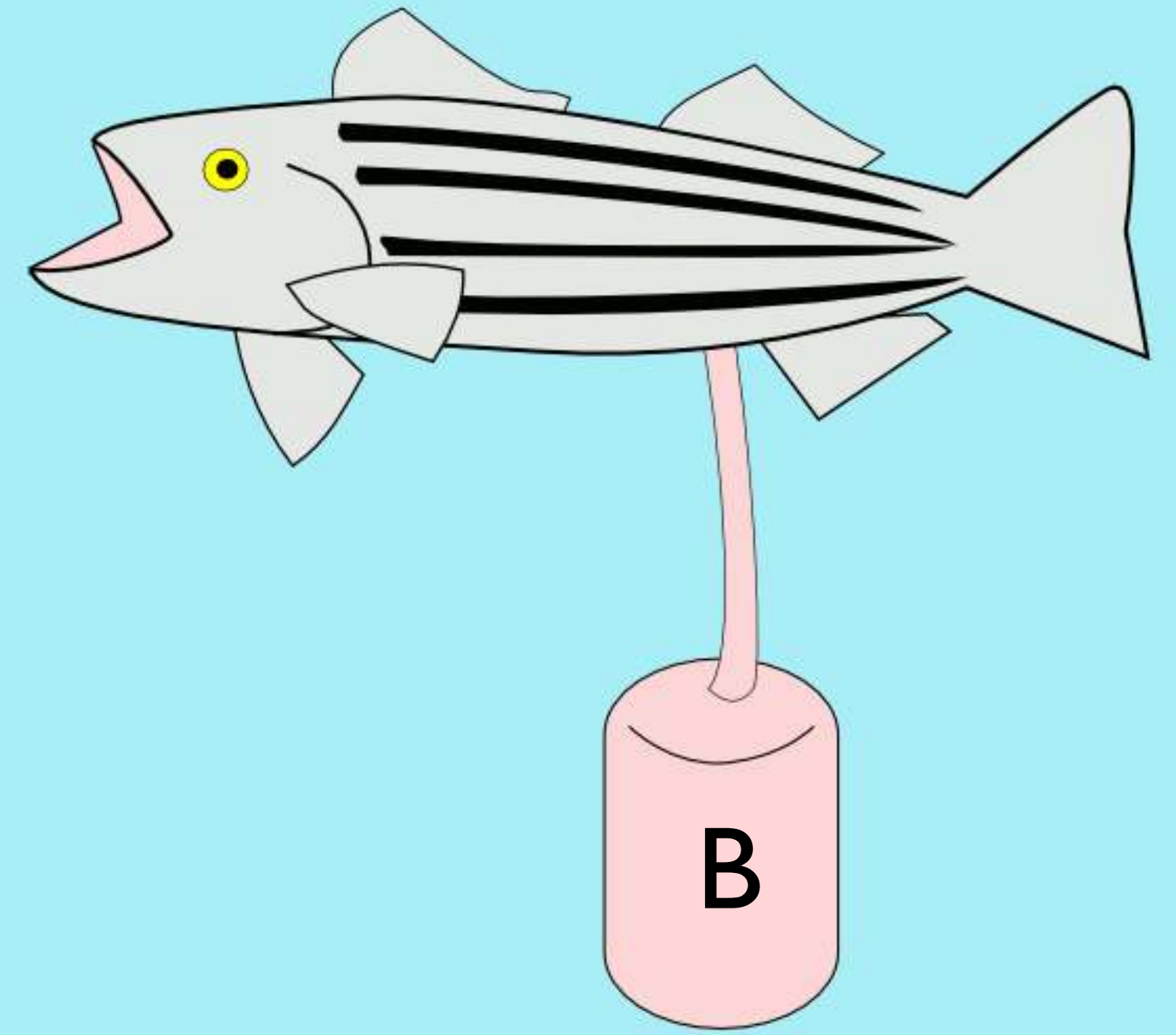
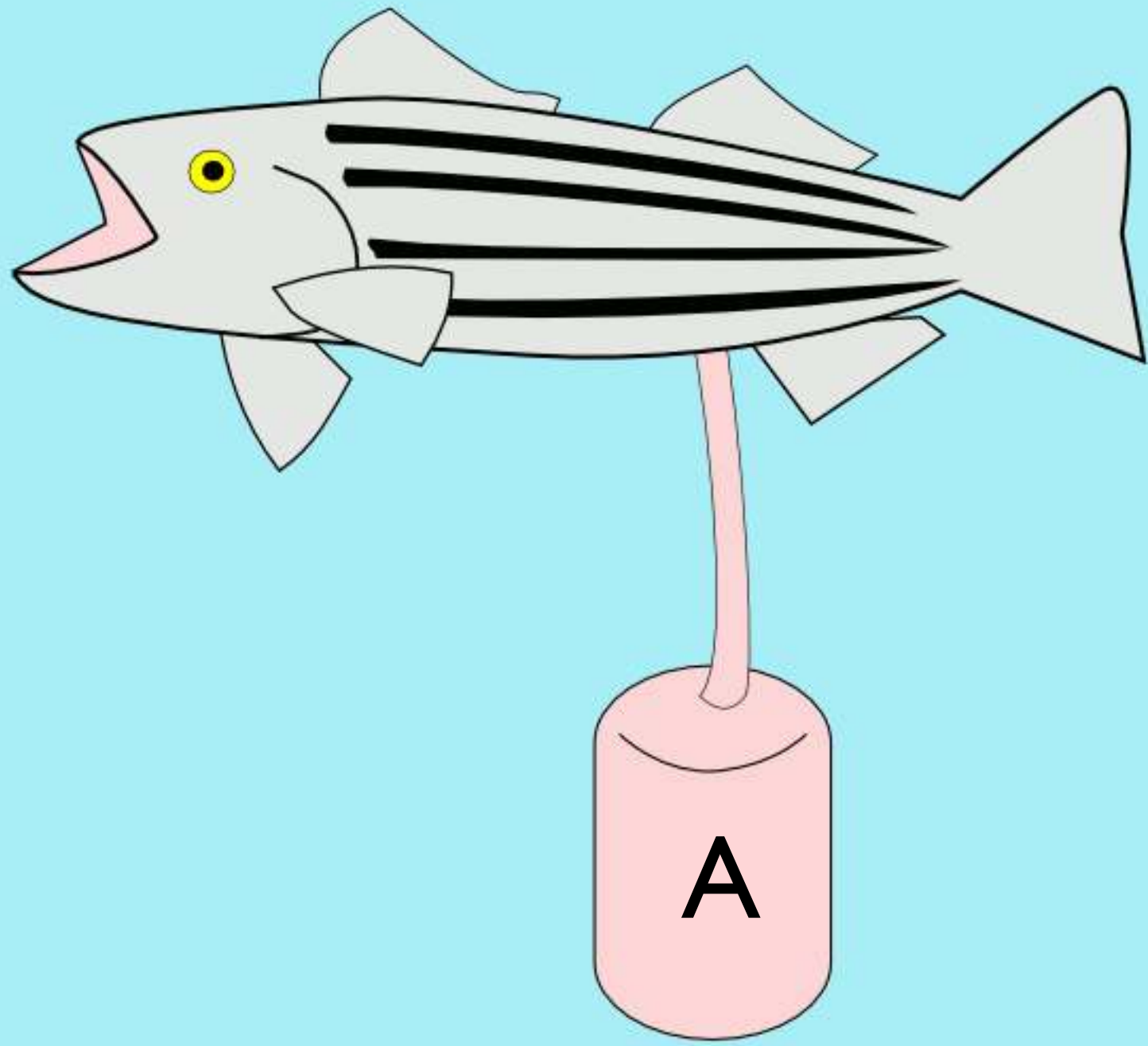
HAIRDRYER

HAIRDRYER

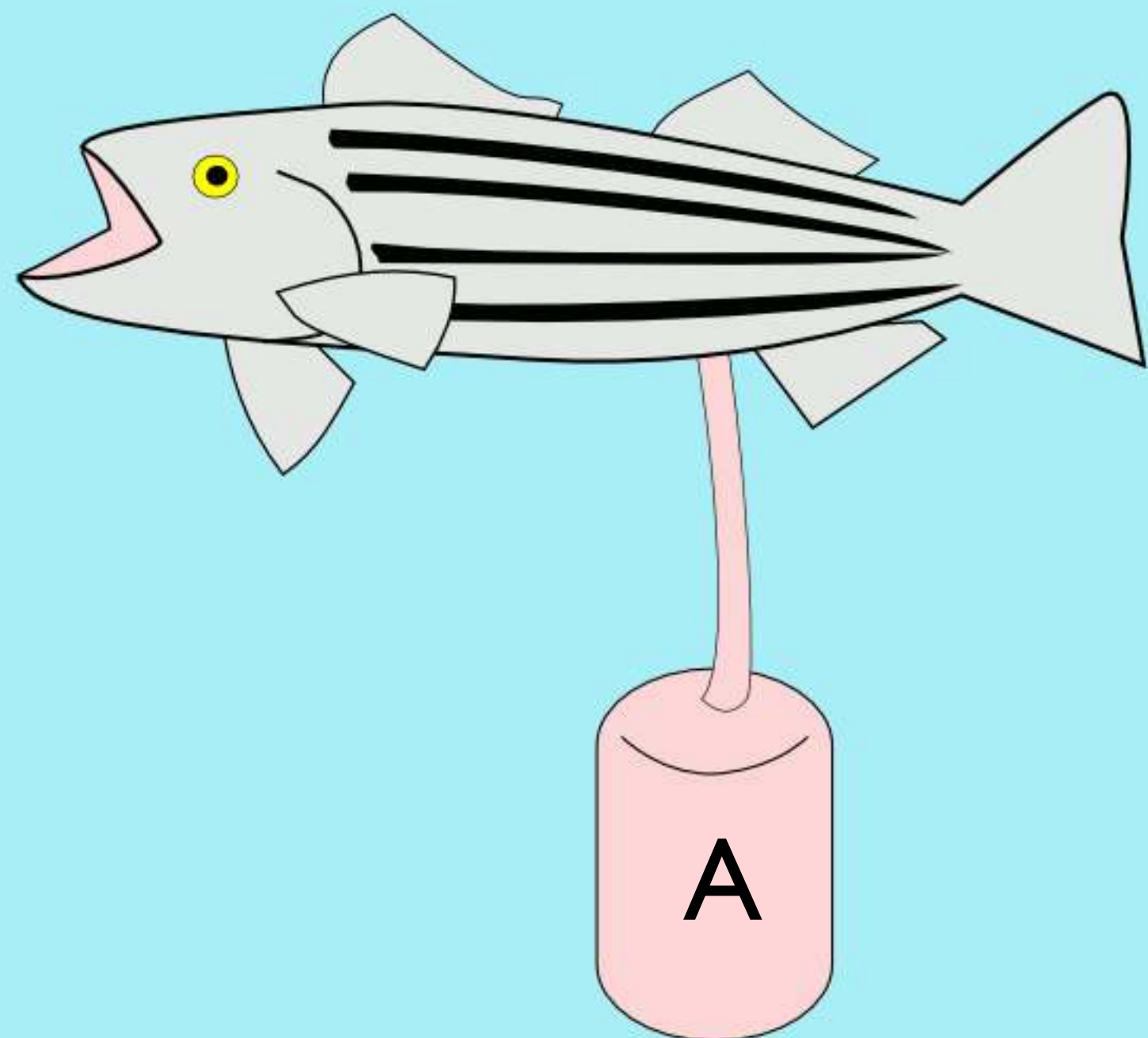
A



B

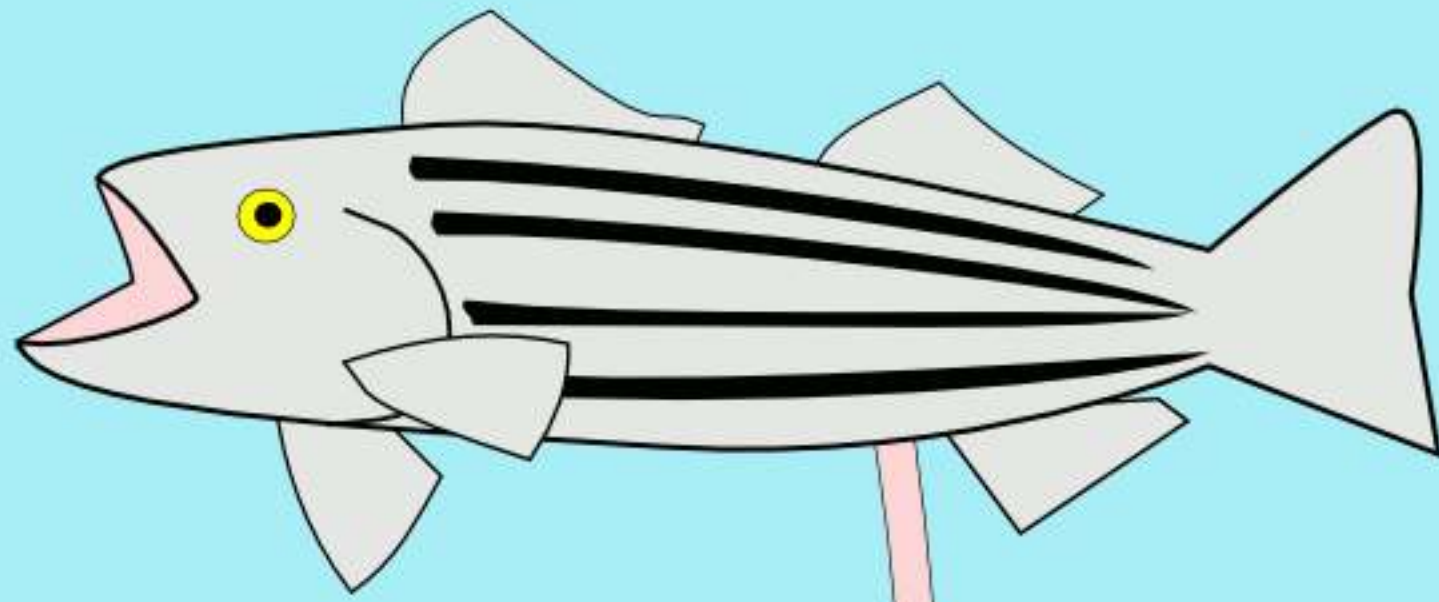


PENCIL CASE

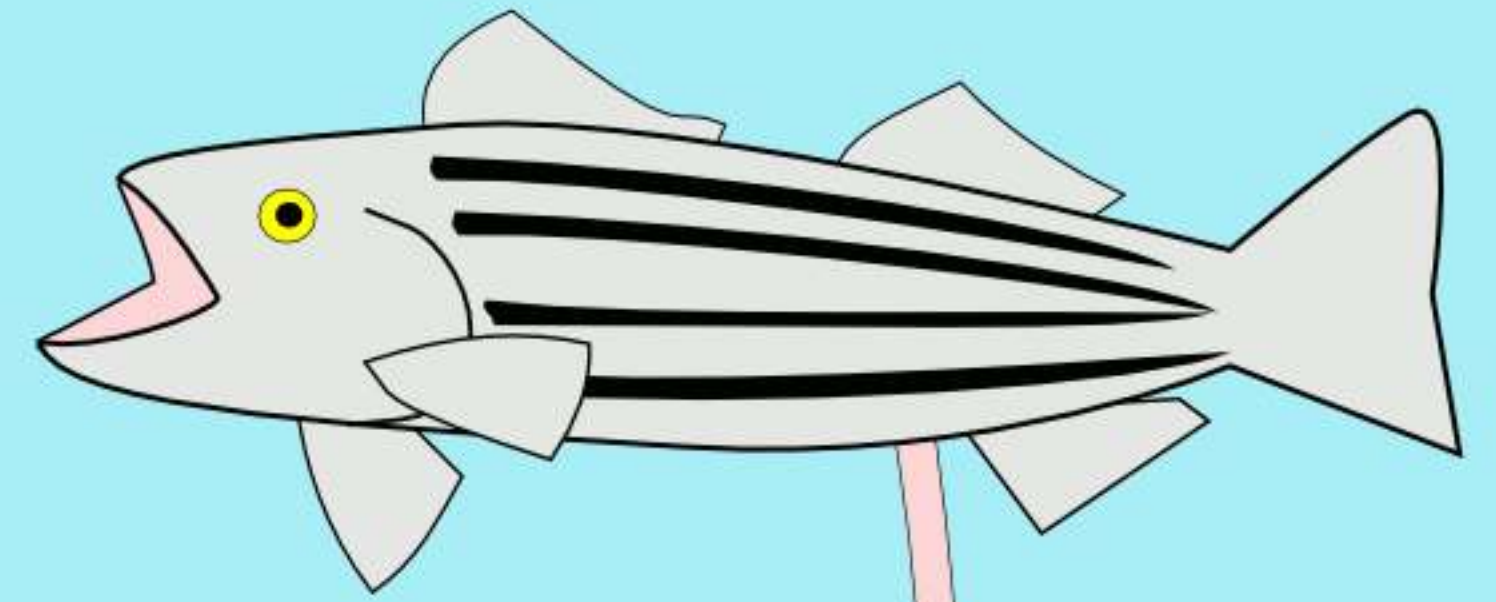


PENCIL CASE





A



B

[HAIRDRYER], [PENCIL CASE]

MERGE

Set Union of Values
Simples

MERGE

Deterministic

MERGE

Deterministic
Idempotent

MERGE

Deterministic
Idempotent
Associative

MERGE

Deterministic

Idempotent

Associative

Commutative

REMOVES?

Set Union?

Absence

How can you tell if X is missing from A and not B because A hasn't seen the addition, or if A has removed X ?

Complexity

The more complex the data, the
harder to model.

Ad Hoc

For each new domain start over

CRDTs

CRDTs

Convergent Replicated Data Types

CRDTs

Convergent Replicated Data Types

CRDTs

Commutative Replicated Data Types

CRDTs

Conflict-free data structures

The Theory



INSTITUT NATIONAL DE RECHERCHE EN INFORMATIQUE ET EN AUTOMATIQUE

A comprehensive study of Convergent and Commutative Replicated Data Types

Marc Shapiro, INRIA & LIP6, Paris, France

Nuno Preguiça, CITI, Universidade Nova de Lisboa, Portugal

Carlos Baquero, Universidade do Minho, Portugal

Marek Zawirski, INRIA & UPMC, Paris, France



INSTITUT NATIONAL DE RECHERCHE EN INFORMATIQUE ET EN AUTOMATIQUE

An Optimized Conflict-free Replicated Set

Annette Bieniusa, INRIA & UPMC, Paris, France

Marek Zawirski, INRIA & UPMC, Paris, France

Nuno Preguiça, CITI, Universidade Nova de Lisboa, Portugal

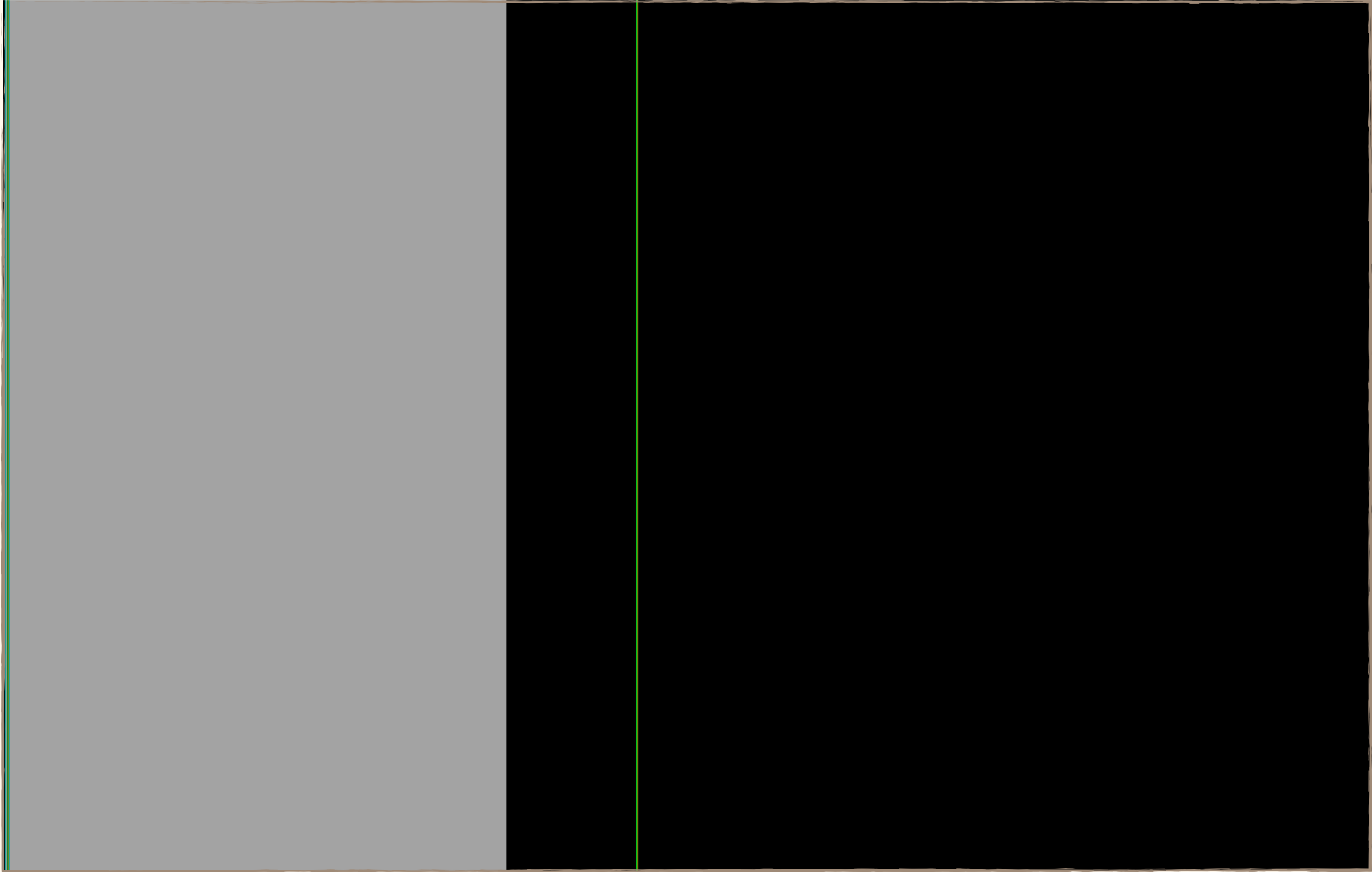
Marc Shapiro, INRIA & LIP6, Paris, France

Carlos Baquero, HASLab, INESC TEC & Universidade do Minho, Portugal

Valter Balegas, CITI, Universidade Nova de Lisboa, Portugal

Sérgio Duarte CITI, Universidade Nova de Lisboa, Portugal

11 Oct 2012



Bounded Join Semilattices

Bounded Join Semilattices

Partially ordered set; least upper bound; ACI.

Bounded Join Semilattices

Associativity: $(X \cdot Y) \cdot Z = X \cdot (Y \cdot Z)$

Bounded Join Semilattices

Commutativity: $X \cdot Y = Y \cdot X$

Bounded Join Semilattices

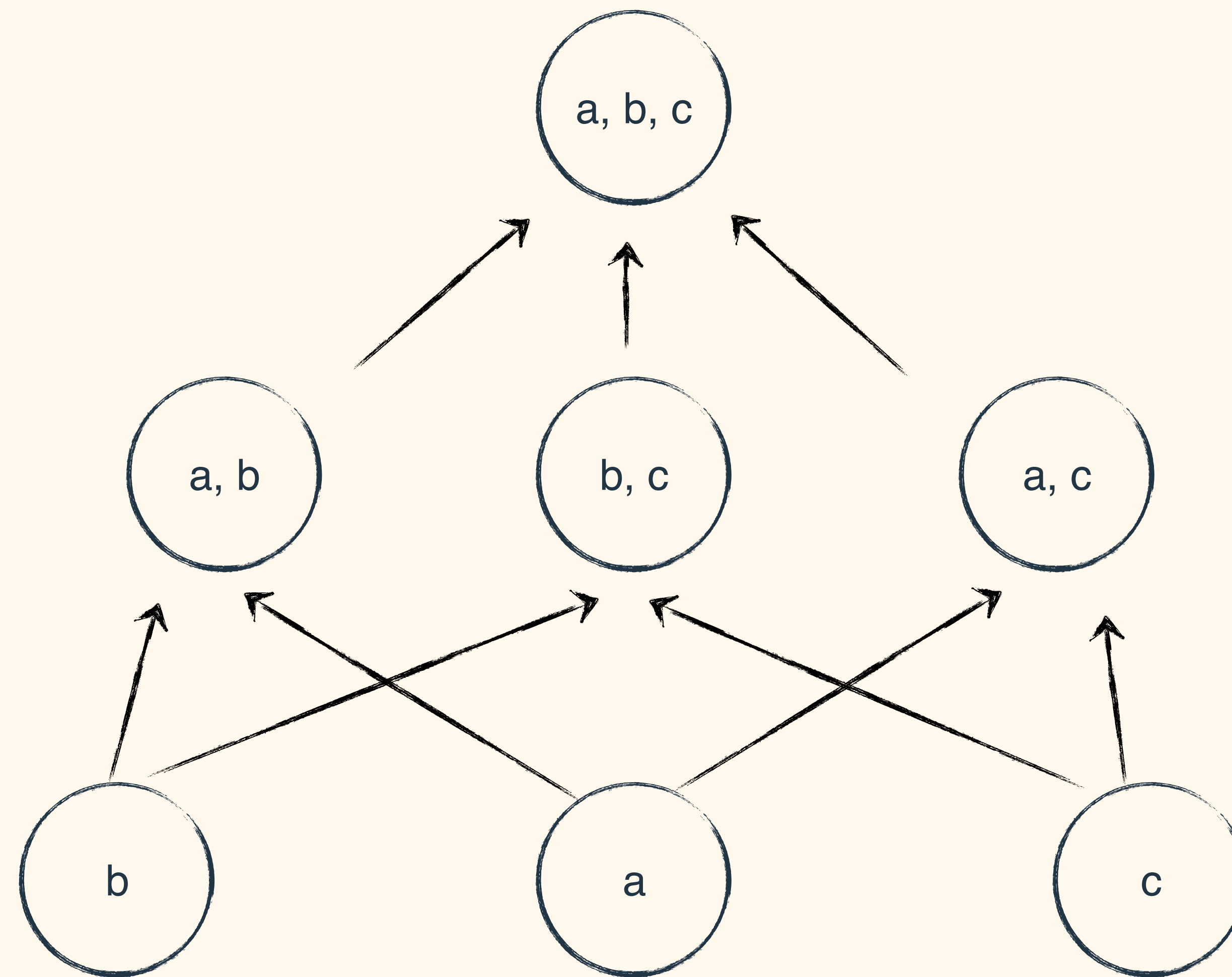
Idempotence: $X \cdot X = X$

Bounded Join Semilattices

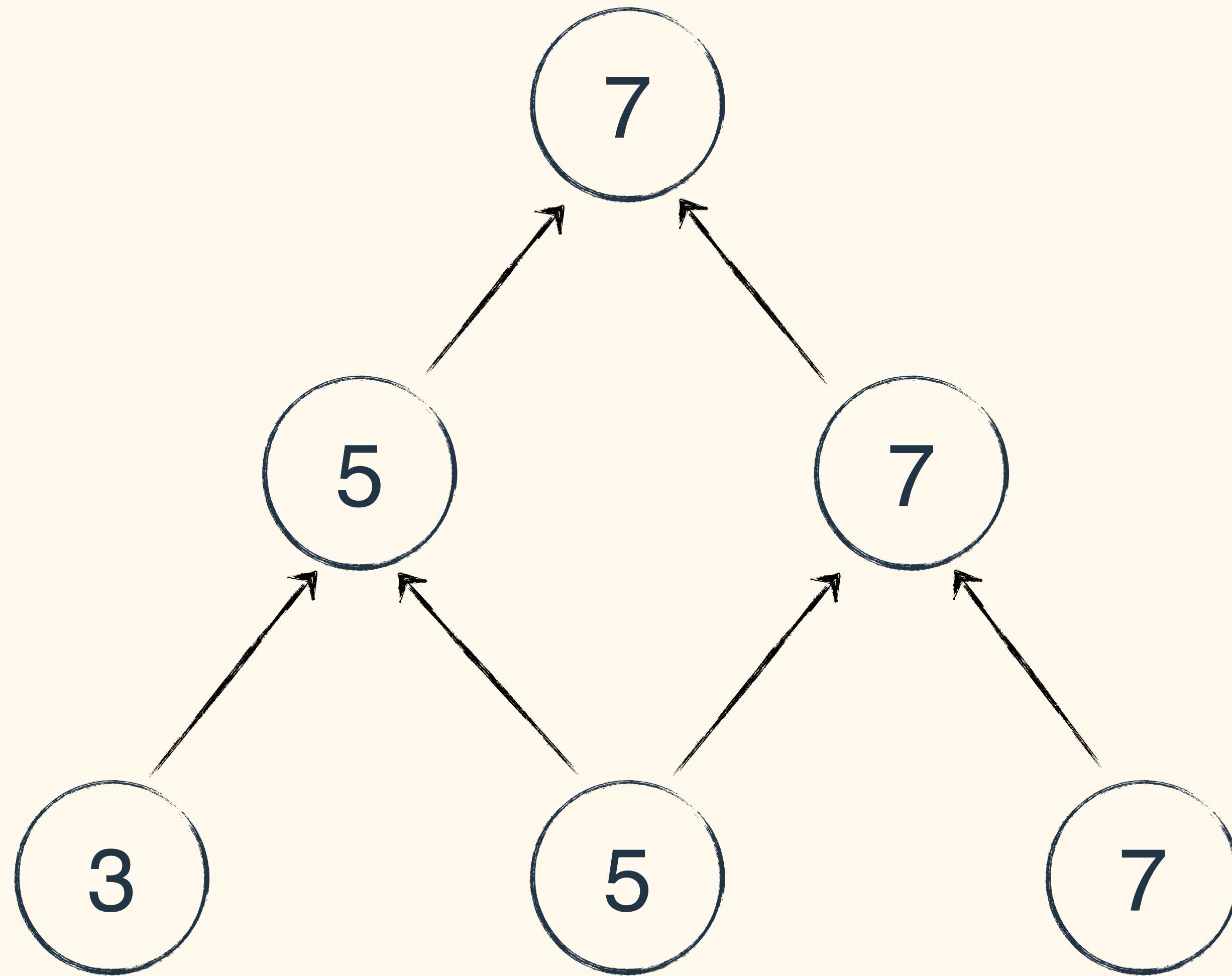
Objects *grow* over time; merge computes LUB

Bounded Join Semilattices

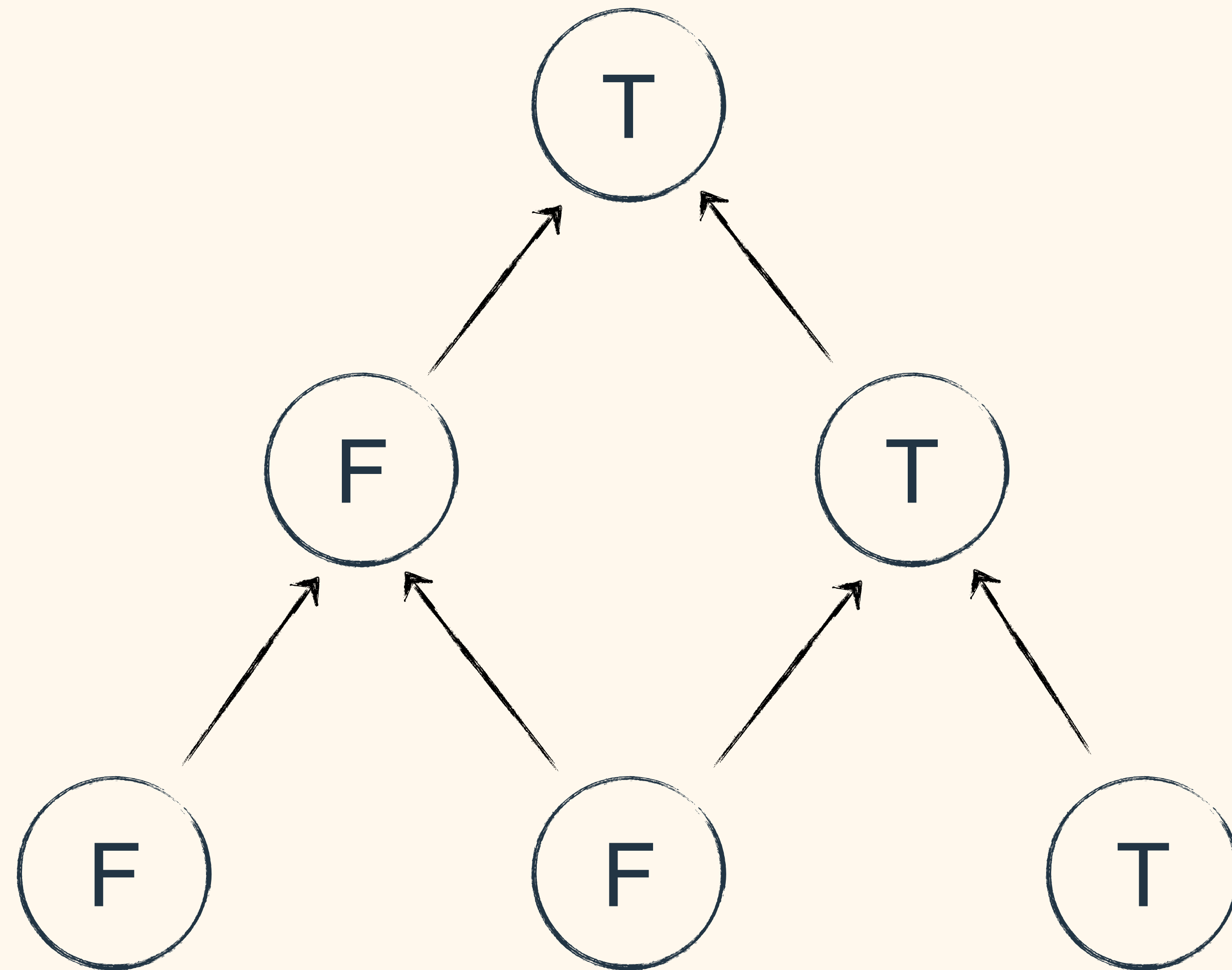
Examples



Set; merge function: **union**.



Increasing natural; merge function: **max**.



Booleans; merge function: **or**.

MERGE!

Deterministic

Idempotent

Associative

Commutative

Riak 2.0

Same Same

But Different

Pick Your Semantic

Add Wins

Remove Wins

Keep Both

Trade Offs

More metadata == bigger objects

ACTORS?

Version Vectors

Entry Per Actor (Charron-Bost)

Riak 1.4

Counters: non-idempotent; $O(\text{Actors})$

Riak 2.0

Sets: Add, Remove, Membership; Idempotent

Riak 2.0

Sets: Add wins; $O(\text{Actors} + \text{Elements})$

Riak 2.0

Maps: Recursive; Associative Array; Nestable

Riak 2.0

Maps: Update wins; $O(\text{Actors} + \text{Elements})$

Riak 2.0

Maps: LWW-Register, Booleans, Sets, and **Maps**

Riak 2.0

LWW-Register: last writer wins

Riak 2.0

Boolean: Enabled, Disabled; $O(\text{Actors})$

riak_dt

```
git clone git@github.com:basho/riak_dt.git
```

EVOLUTION

OF A SET

Causality

Version Vectors

$[\{a, 1\}, \{b, 3\}, \{c, 2\}]$

Causality

Version Vectors

$[\{a, 2\}, \{b, 3\}, \{c, 2\}]$

$>$

$[\{a, 1\}, \{b, 3\}, \{c, 2\}]$

Causality

Version Vectors

$[\{a, 2\}, \{b, 3\}, \{c, 2\}]$

$[\{a, 1\}, \{b, 4\}, \{c, 2\}]$

$[\{a, 2\}, \{d, 1\}, \{c, 2\}]$

$[\{a, 2\}, \{b, 4\}, \{c, 2\}]$

Causality

Version Vectors

“Dots” are Events

Causality

“Dots” are Events

$[\{a, 2\}, \{b, 3\}, \{c, 2\}]$

$\{b, 1\}$

$\{b, 2\}$

$\{b, 3\}$

Evolution of a Set

- GSET

Evolution of a Set

- GSET

Evolution of a Set

- GSET
- 2P-Set

Evolution of a Set

- U-Set

Evolution of a Set

- U-Set
- OR-Set

Evolution of a Set

- U-Set
- OR-Set

Evolution of a Set

- U-Set
- OR-Set
- ORSWOT

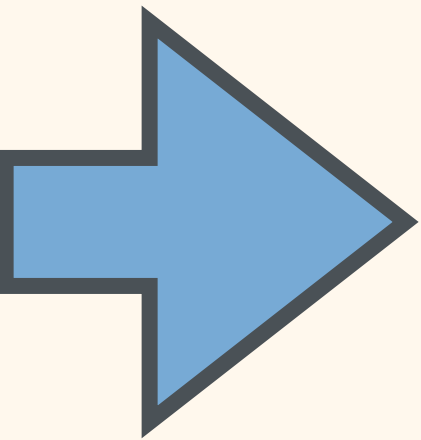
$\{a, 1\}$

$\{a, 1\}$

Shelly

$\{\text{a}, 1\}$

$\{\text{a}, 1\}$	Shelly
-------------------	--------



$\{\text{a}, 1\}$

$\{\text{a}, 1\}$	Shelly
-------------------	--------

$\{\{a, 1\}\}$

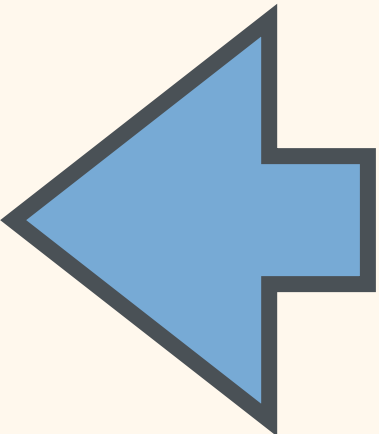
$\{a, 1\}$	Shelly
------------	--------

$\{\{a, 1\}, \{b, 3\}\}$

$\{a, 1\}$	Shelly
$\{b, 1\}$	Bob
$\{b, 2\}$	Phil
$\{b, 3\}$	Pete

$[\{a, 1\}, \{b, 3\}]$

$\{a, 1\}$	Shelly
$\{b, 1\}$	Bob
$\{b, 2\}$	Phil
$\{b, 3\}$	Pete



$[\{a, 1\}, \{b, 3\}]$

$\{a, 1\}$	Shelly
$\{b, 1\}$	Bob
$\{b, 2\}$	Phil
$\{b, 3\}$	Pete

$[\{a, 2\}, \{b, 3\}]$

$\{a, 1\}$	Shelly
$\{b, 1\}$	Bob
$\{b, 3\}$	Pete
$\{a, 2\}$	Anna

$[\{a, 1\}, \{b, 3\}]$

$\{a, 1\}$	Shelly
$\{b, 1\}$	Bob
$\{b, 2\}$	Phil
$\{b, 3\}$	Pete

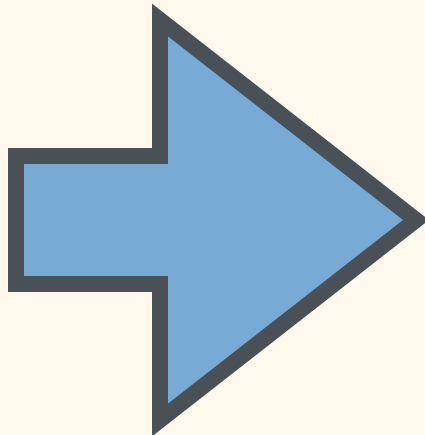
$[\{a, 2\}, \{b, 3\}]$

$\{a, 1\}$	Shelly
------------	--------

$\{b, 1\}$	Bob
------------	-----

$\{b, 3\}$	Pete
------------	------

$\{a, 2\}$	Anna
------------	------



$[\{a, 2\}, \{b, 3\}]$

$\{a, 1\}$	Shelly
------------	--------

$\{b, 1\}$	Bob
------------	-----

$\{b, 3\}$	Pete
------------	------

$\{a, 2\}$	Anna
------------	------

GARBAGE

Tombstones
Dead Actors

GARBAGE

~~Tombstones~~

Dead Actors

No More Siblings

But that is all

MORE?



This project is funded by the European Union,
7th Research Framework Programme, ICT call 10,
grant agreement n°609551.



Questions?